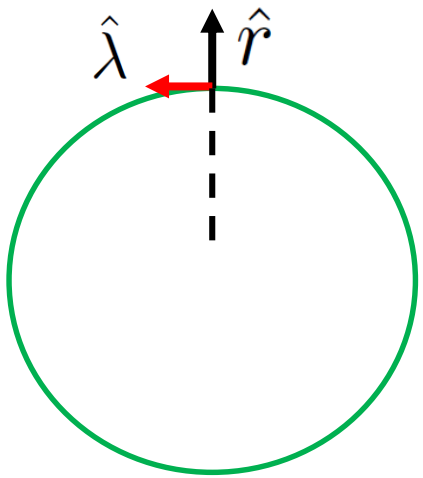
圓柱座標方程式推導 Cylindrical Coordinate

$$u(t,r,\lambda,z) \quad v(t,r,\lambda,z)$$



$$\frac{D}{Dt} = \frac{\partial}{\partial t}\frac{Dt}{Dt} + \frac{\partial}{\partial r}\frac{Dr}{Dt} + \frac{\partial}{\partial \lambda}\frac{D\lambda}{Dt} + \frac{\partial}{\partial z}\frac{Dz}{Dt}$$

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \frac{Dr}{Dt}\frac{\partial}{\partial r} + r\frac{D\lambda}{Dt}\frac{\partial}{r\partial \lambda} + \frac{Dz}{Dt}\frac{\partial}{\partial z}$$

$$\boxed{\begin{gathered} u = \frac{Dr}{Dt} \quad v = r\frac{D\lambda}{Dt} \quad w = \frac{Dz}{Dt} \\ \frac{D}{Dt} = \frac{\partial}{\partial t} + u\frac{\partial}{\partial r} + v\frac{\partial}{r\partial \lambda} + w\frac{\partial}{\partial z} \end{gathered}}$$

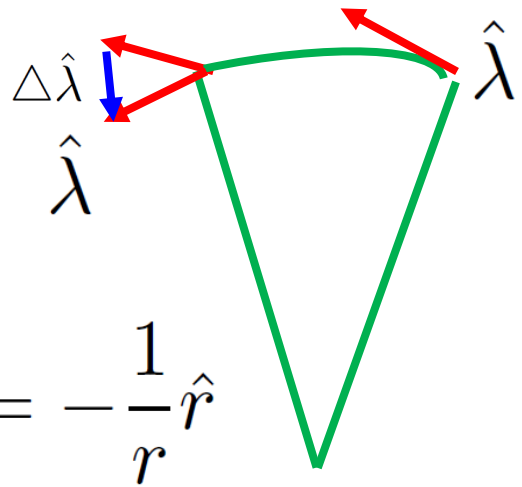
$$\vec{V} = u\hat{r} + v\hat{\lambda}$$

$$\frac{D\vec{V}}{Dt} + f\hat{k} \times \vec{V} = -\frac{1}{\rho}\nabla p$$

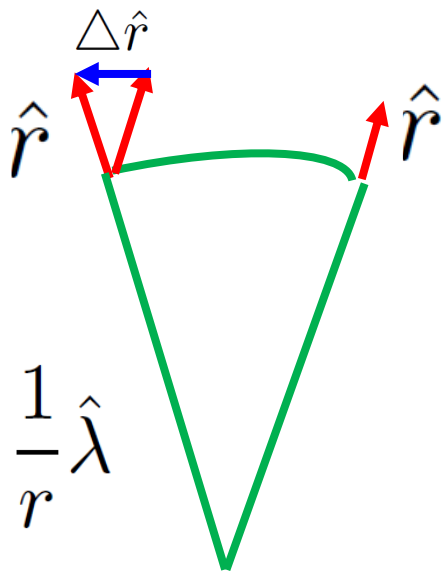
$$\hat{k} \times \hat{r} = \hat{\lambda} \qquad \hat{k} \times \hat{\lambda} = -\hat{r}$$

$$\frac{D\vec{V}}{Dt} = \frac{Du}{Dt}\hat{r} + u\frac{D\hat{r}}{Dt} + \frac{Dv}{Dt}\hat{\lambda} + v\frac{D\hat{\lambda}}{Dt}$$

$$\boxed{\begin{aligned} \frac{D\hat{r}}{Dt} &= v\frac{\partial\hat{r}}{r\partial\lambda} = \frac{v}{r}\frac{\partial\hat{r}}{\partial\lambda} = \frac{v}{r}\hat{\lambda} \\ \frac{D\hat{\lambda}}{Dt} &= v\frac{\partial\hat{\lambda}}{r\partial\lambda} = \frac{v}{r}\frac{\partial\hat{\lambda}}{\partial\lambda} = -\frac{v}{r}\hat{r} \end{aligned}}$$



$$\frac{\triangle\hat{\lambda}}{r\triangle\lambda} = -\frac{1}{r}\hat{r}$$



$$\frac{\triangle\hat{r}}{r\triangle\lambda} = \frac{1}{r}\hat{\lambda}$$

$$\frac{Dv}{Dt} + \frac{uv}{r} + fu = -\frac{1}{\rho}\frac{\partial p}{r\partial\lambda} \qquad \hat{\lambda}$$

$$\frac{Du}{Dt} - \frac{v^2}{r} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial r} \qquad \hat{r}$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial r} + v\frac{\partial v}{r\partial\lambda} + w\frac{\partial v}{\partial z} + \frac{uv}{r} + fu = -\frac{1}{\rho}\frac{\partial p}{r\partial\lambda}$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial r} + v\frac{\partial u}{r\partial\lambda} + w\frac{\partial u}{\partial z} - \frac{v^2}{r} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial r}$$

$$\frac{\partial}{\partial \lambda} = 0$$ axisymmetric

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial r} + w\frac{\partial v}{\partial z} + \frac{uv}{r} + fu = 0$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial r} + w\frac{\partial u}{\partial z} - \frac{v^2}{r} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial r}$$

$$\frac{\partial}{\partial z} = 0$$ + Axisymmetric 2D

$$\frac{\partial v}{\partial t} = -u\left(f + \zeta\right)$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial r} - \frac{v^2}{r} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial r}$$

$$\frac{\partial v}{\partial t} + u\left(f + \frac{\partial v}{\partial r} + \frac{v}{r}\right) + w\frac{\partial v}{\partial z} = 0$$

$$\frac{\partial v}{\partial t} = -u\left(f + \zeta\right) - w\frac{\partial v}{\partial z}$$

Gradient wind balance

$$\frac{v^2}{r} + fv = \frac{1}{\rho}\frac{\partial p}{\partial r}$$

$$\zeta = \frac{\partial(rv)}{r\partial r} \qquad D = \frac{\partial(ru)}{r\partial r}$$

$$\frac{\partial v}{\partial t} = -u\left(f + \zeta\right)$$

$$\frac{\partial}{r\partial r}\left(r\frac{\partial v}{\partial t} = -ru\left(f + \zeta\right)\right)$$

$$\frac{\partial}{\partial t}\left(\frac{\partial(rv)}{r\partial r}\right) = -\frac{\partial(ru)}{r\partial r}\left(f + \zeta\right) - u\frac{\partial}{\partial r}\left(f + \zeta\right)$$

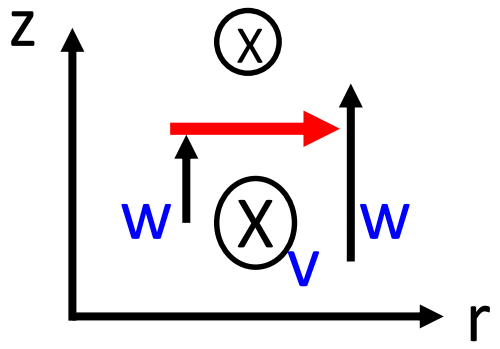
$$\frac{\partial \zeta}{\partial t} + u\frac{\partial \zeta}{\partial r} = -\frac{\partial(ru)}{r\partial r}\left(f + \zeta\right)$$

$$\frac{D\zeta}{Dt} = -\frac{\partial(ru)}{r\partial r}\left(f + \zeta\right) = -D\left(f + \zeta\right)$$

$$\frac{\partial v}{\partial t} = -u\left(f+\zeta\right) - w\frac{\partial v}{\partial z}$$

$$-\frac{\partial rw}{r\partial r}\frac{\partial v}{\partial z} = -\frac{\partial w}{\partial r}\frac{\partial v}{\partial z} - w\frac{\partial}{\partial z}(\frac{v}{r})$$

$$\frac{\partial}{r\partial r}\left(r\frac{\partial v}{\partial t} = -ru\left(f+\zeta\right) - rw\frac{\partial v}{\partial z})\right)$$



$$\frac{D\zeta}{Dt} = \frac{\partial\zeta}{\partial t} + u\frac{\partial\zeta}{\partial r} + w\frac{\partial\zeta}{\partial z} = -\frac{\partial(ru)}{r\partial r}\left(f+\zeta\right) - \frac{\partial rw}{r\partial r}\frac{\partial v}{\partial z}$$

$$\frac{D\zeta}{Dt} = -\frac{\partial(ru)}{r\partial r}\left(f+\zeta\right) - \frac{\partial rw}{r\partial r}\frac{\partial v}{\partial z}$$

$$\frac{\partial v}{\partial z} < 0 \qquad \frac{\partial}{\partial z}(\frac{v}{r}) < 0$$

$$-\frac{\partial w}{\partial r}\frac{\partial v}{\partial z} > 0 \qquad -w\frac{\partial}{\partial z}(\frac{v}{r}) > 0$$

梯度風旱地轉風關係

$$R_o = \frac{v^2/r}{fv} = \frac{v}{fr}$$

> 1 渦旋

< 1 QG

$$\frac{v^2}{r} + fv = \frac{1}{\rho}\frac{\partial p}{\partial r} = fv_g$$

Gradient wind balance

$$v^2 + rfv - rfv_g = 0$$

$$v = \frac{-rf \pm \sqrt{(rf)^2 + 4rfv_g}}{2}$$

$$v = \frac{-rf \pm rf\sqrt{1 + 4v_g/(rf)}}{2}$$

$$\frac{v_g}{rf} << 1$$

V_g 很小 (壓力梯度很小 熱帶)

或rf 很大 (流場趨近直線曲率很大
或科是參數很大 中緯度)

$$v \approx \frac{-rf \pm rf\,(1 + 2v_g/(rf))}{2}$$

Negative root $v \approx -rf - v_g \approx -rf$

Positive root $v \approx v_g$

圓柱座標次環流方程式推導

$$\frac{v^2}{r} + fv = \frac{\partial \phi}{\partial r}$$

$$\frac{\partial v}{\partial t} + u\left(f + \zeta\right) + w\frac{\partial v}{\partial z} = \mathbf{X}$$

$$\frac{\partial \phi}{\partial z} = \frac{g}{\theta_0}\theta$$

$$\frac{\partial \rho r u}{r \partial r} + \frac{\partial \rho w}{\partial z} = 0$$

$$\frac{\partial \theta}{\partial t} + u\frac{\partial \theta}{\partial r} + w\frac{\partial \theta}{\partial z} = \frac{\theta}{T}\frac{Q}{c_p}$$

Vortex Structure Parameter

$$\rho A = \frac{g}{\theta_0}\frac{\partial \theta}{\partial z}$$

$$\rho B = -\frac{g}{\theta_0}\frac{\partial \theta}{\partial r} = -\left(f + \frac{2v}{r}\right)\frac{\partial v}{\partial z}$$

$$\rho C = \left(f + \frac{2v}{r}\right)\left(f + \frac{\partial r v}{r \partial r}\right)$$

$$\rho w = \frac{\partial r \psi}{r \partial r} \qquad \rho u = -\frac{\partial \psi}{\partial z}$$

$$\zeta = \frac{\partial (r v)}{r \partial r}$$

$$(f+\frac{2v}{r})\left(\frac{\partial v}{\partial t}+u\,(f+\zeta)+w\frac{\partial v}{\partial z}=\mathbf{X}\right)$$

$$\frac{\partial}{\partial t}(fv+\frac{v^2}{r})+u\,(f+\zeta)\,(f+\frac{2v}{r})+w\frac{\partial v}{\partial z}(f+\frac{2v}{r})=(f+\frac{2v}{r})\mathbf{X}=F$$

主環流以熱力風平衡
緩慢變化

$$\frac{g}{\theta_0}\left(\frac{\partial\theta}{\partial t}+u\frac{\partial\theta}{\partial r}+w\frac{\partial\theta}{\partial z}=\frac{\theta}{T}\frac{Q}{c_p}\right)$$

摩擦與對流外力
破壞熱力風平衡 $\frac{\partial J}{\partial r}-\frac{\partial F}{\partial z}$

$$\frac{\partial^2\phi}{\partial t\partial z}-\rho Bu+\rho Aw=\frac{g}{\theta_0}\frac{\theta}{T}Q=J \qquad \frac{\partial^2\phi}{\partial t\partial r}+\rho Cu-\rho Bw=F$$

次環流調節使得熱力風平衡能維持

$$\rho w=\frac{\partial r\psi}{r\partial r} \qquad \rho u=-\frac{\partial\psi}{\partial z}$$

$$\mathcal{L}(\ )=\frac{\partial}{\partial r}\left(A\frac{\partial r(\ )}{r\partial r}+B\frac{\partial(\ )}{\partial z}\right)+\frac{\partial}{\partial z}\left(B\frac{\partial r(\ )}{r\partial r}+C\frac{\partial(\ )}{\partial z}\right)$$

$$\mathcal{L}\psi=\frac{\partial J}{\partial r}-\frac{\partial F}{\partial z}$$

$$\int\int\chi\,\mathcal{L}\,\psi\,rdrdz=\int\int\psi\,\mathcal{L}\,\chi\,rdrdz$$

次環流方程式
2nd circulation equation

Symmetric overturning circulation

$$f_0 U = -\frac{\partial \bar{\phi}}{\partial y} \qquad f_0 \frac{\partial U}{\partial z} = -\frac{\partial \bar{b}}{\partial z}$$

基本場 basic state

$$U = U(y,z) \quad \bar{b} = \bar{b}(y,z) \quad \frac{\partial}{\partial x} = 0$$

$$u' = u'(x,y,z,t) \quad b' = b'(x,y,z,t)$$

Primary circulation equations

$$\frac{\partial u'}{\partial t} + v'\frac{\partial U}{\partial y} + w'\frac{\partial U}{\partial z} - f_0 v' = \frac{F}{f_0}$$

$$\frac{\partial b'}{\partial t} + v'\frac{\partial \bar{b}}{\partial y} + w'\frac{\partial \bar{b}}{\partial z} = Q$$

$$N^2 = \frac{\partial \bar{b}}{\partial z} = A \qquad f_0 \frac{\partial U}{\partial z} = -\frac{\partial \bar{b}}{\partial y} = B \qquad f_0\left(f_0 - \frac{\partial U}{\partial y}\right) = C$$

基本場 basic state

$$AC - B^2 = f_0\left(f_0 - \frac{\partial U}{\partial y}\right)\frac{\partial \bar{b}}{\partial z} + f_0 \frac{\partial U}{\partial z}\frac{\partial \bar{b}}{\partial y} = f_0 P$$

Secondary circulation equations

$$\frac{\partial v'}{\partial t} + f_0 u' = -\frac{\partial \phi'}{\partial y}$$

$$\frac{\partial w'}{\partial t} = -\frac{\partial \phi'}{\partial z} + b'$$

$$\frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0$$

$$\frac{\partial}{\partial t}\left(\frac{\partial w'}{\partial y} - \frac{\partial v'}{\partial z}\right) = \frac{\partial b'}{\partial y} + \frac{\partial f_0 u'}{\partial z}$$

$$w' = \frac{\partial \psi}{\partial y} \qquad v' = -\frac{\partial \psi}{\partial z}$$

直角座標次環流方程式推導
物理意義

$$\frac{\partial f_0 u'}{\partial t} = f_0 \left(f_0 - \frac{\partial U}{\partial y} \right) v' - w' \frac{\partial U}{\partial z} + F$$

$$w' = \frac{\partial \psi}{\partial y} \qquad v' = -\frac{\partial \psi}{\partial z}$$

$$\frac{\partial f_0 u'}{\partial t} = C v' - B w' + F$$

$$\frac{\partial b'}{\partial t} = B v' - A w' + Q$$

Primary circulation equations

$$\frac{\partial^2}{\partial t^2} \nabla^2 \psi + \frac{\partial}{\partial y} \left(A \frac{\partial \psi}{\partial y} + B \frac{\partial \psi}{\partial z} \right) + \frac{\partial}{\partial z} \left(B \frac{\partial \psi}{\partial y} + C \frac{\partial \psi}{\partial z} \right) = \frac{\partial Q}{\partial y} + \frac{\partial F}{\partial z}$$

Secondary circulation equations

$$\mathcal{L} = \frac{\partial}{\partial y} \left(A \frac{\partial}{\partial y} + B \frac{\partial}{\partial z} \right) + \frac{\partial}{\partial z} \left(B \frac{\partial}{\partial y} + C \frac{\partial}{\partial z} \right)$$

$$\frac{\partial^2}{\partial t^2} \nabla^2 \psi + \mathcal{L} \psi = \frac{\partial Q}{\partial y} + \frac{\partial F}{\partial z}$$

Eliassen transverse circulation equations

$$\mathcal{L} \psi = \frac{\partial Q}{\partial y} + \frac{\partial F}{\partial z}$$

波動分析

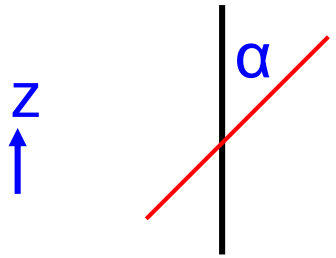
$$e^{iky+imz-i\omega t}$$

$$\frac{\partial^2}{\partial t^2}\nabla^2\psi + A\frac{\partial^2\psi}{\partial y^2} + 2B\frac{\partial^2\psi}{\partial y\partial z} + C\frac{\partial^2\psi}{\partial z^2} = 0$$

$$\omega^2 = A\cos^2\alpha + 2B\cos\alpha\sin\alpha + C\sin^2\alpha$$

U = 0 B = 0

$$\omega^2 = \frac{k^2N^2 + m^2f^2}{k^2+m^2} = N^2\cos^2\alpha + f_0^2\sin^2\alpha$$



$$f_0 y - U = M$$

$$\omega^2 = A\cos^2\alpha + 2B\cos\alpha\sin\alpha + C\sin^2\alpha$$

$$= A\left(\frac{1+\cos 2\alpha}{2}\right) + B\sin 2\alpha + C\left(\frac{1-\cos 2\alpha}{2}\right)$$

$$= \frac{A+C}{2} + \left(\frac{A-C}{2}\right)\cos 2\alpha + B\sin 2\alpha$$

$$\omega^2_{max} = \frac{A+C}{2} + W$$

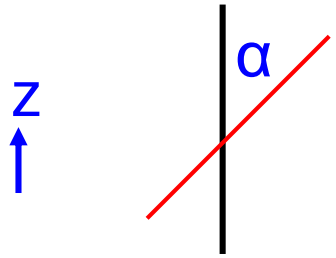
$$\omega^2_{min} = \frac{A+C}{2} - W$$

$$W^2 = \left(\frac{A-C}{2}\right)^2 + B^2$$

$$\omega^2_{max}\omega^2_{min} = \frac{(A+C)^2}{4} - \frac{(A-C)^2}{4} - B^2$$

$$= AC - B^2 = J(\bar{M},\bar{b}) = P$$

PV conservation indicates stiffness conserved!

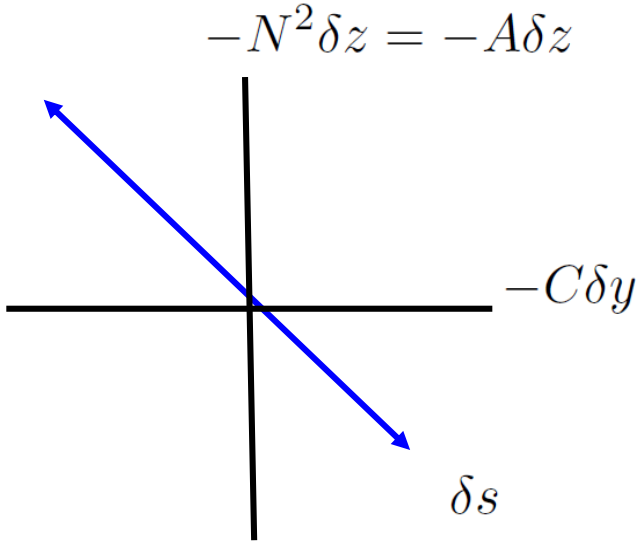


U = 0 B = 0

$$\omega^2 = \frac{k^2 N^2 + m^2 f^2}{k^2 + m^2} = N^2 \cos^2 \alpha + f_0^2 \sin^2 \alpha$$

$$\frac{d^2 \delta s}{dt^2} = \left(-A \cos^2 \alpha - 2B \cos \alpha \sin \alpha - C \sin^2 \alpha\right) \delta s$$

$$\omega^2 = A \cos^2 \alpha + 2B \cos \alpha \sin \alpha + C \sin^2 \alpha$$



U = 0 B = 0

$$\frac{d^2 \delta s}{dt^2} = -A \delta z \cos \alpha - C \delta y \sin \alpha$$

$$= -A \delta s \cos^2 \alpha - C \delta s \sin^2 \alpha$$

$$\frac{d^2 \delta s}{dt^2} = \left(-A \cos^2 \alpha - C \sin^2 \alpha\right) \delta s$$

$$\frac{\omega^2}{\sin^2\alpha} = A\cot^2\alpha + 2B\cot\alpha + C = A\left(\cot\alpha + \frac{B}{A}\right)^2 + \frac{AC - B^2}{A}$$

沿著θ面

$$\frac{\omega^2}{\cos^2\alpha} = A + 2B\tan\alpha + C\tan^2\alpha = C\left(\tan\alpha + \frac{B}{C}\right)^2 + \frac{AC - B^2}{C}$$

沿著M角動量面

$$\omega^2 > 0 \Longleftrightarrow AC - B^2 = PV > 0$$

穩定 stable

$$\omega^2 < 0 \Longrightarrow AC - B^2 = PV < 0$$

不穩定 unstable

必要條件

$$AC - B^2 < 0, \quad \cot\alpha = -\frac{B}{A} \; or \; \tan\alpha = -\frac{B}{C} \Longrightarrow \omega^2 < 0$$

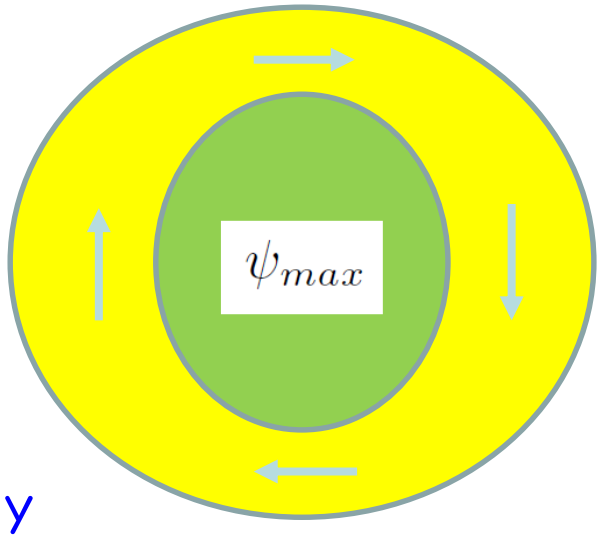
位渦<0 沿著θ面 或 沿著M角動量面

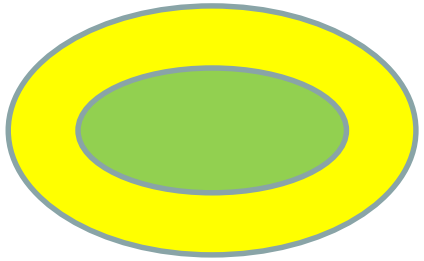
不穩定必要條件

Eliassen Transverse Equation

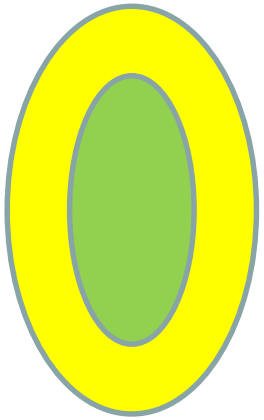
$$\mathcal{L} = \frac{\partial}{\partial y}\left(A\frac{\partial}{\partial y} + B\frac{\partial}{\partial z}\right) + \frac{\partial}{\partial z}\left(B\frac{\partial}{\partial y} + C\frac{\partial}{\partial z}\right)$$

$$\mathcal{L}\psi = \frac{\partial Q}{\partial y} + \frac{\partial F}{\partial z}$$

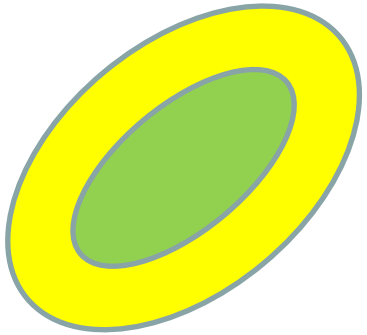




A >> C
垂直穩定度主宰
垂直範圍小
水平拉長 y 微分小



A << C
慣性穩定度主宰
水平被限制
垂直拉長 z y 微分小



B
傾斜項

$$\frac{\partial Q}{\partial y} < 0$$

$$w' = \frac{\partial \psi}{\partial y} \qquad v' = -\frac{\partial \psi}{\partial z}$$

垂直運動絕熱增温(北邊)冷卻(南邊)
減弱Qy < 0效應 (減弱南北温差)

科氏力矩增加西風(上邊)東風(下邊)反應
Qy < 0 的熱力風平衡 (增加垂直風切)

$PV > 0 \quad \omega_{max}\,\omega_{min} > 0$ sufficient condition for symmetric stability

PV is a measure of the "stiffness" of the baroclinic fluid with respect to symmetrical disturbance. Any inviscid, adiabatic rearrangement of the unidirectional flow must preserve the product of the minimum and maximum frequencies for symmetric disturbance.

$PV < 0 \quad \omega_{min} < 0 < \omega_{max}$

During a front-genetic process, the static stability increases in the front, so that ω_{max} increases there. If PV is conserved, then ω_{min} decreases. That is, the stability to vertical motion is increased but that to motion along isentropes is decreased. At this point, it would be possible that turbulent diffusion could destabilize the front either by permitting PV to change sign, or by relaxing the inertial or gravitational constant. However, it is more likely that symmetric instability would be triggered by the effective decrease in static stability due to latent heat release.

Hoskins 1980-

$$P = AC - B^2 < 0$$

$$\left(f_0 - \frac{\partial U}{\partial y}\right)\frac{\partial \bar{b}}{\partial z} < f_0(\frac{\partial U}{\partial z})^2$$

$$R_i = \frac{\partial \bar{b}}{\partial z}/(\frac{\partial U}{\partial z})^2$$

$$R_i < f_0/\left(f_0 - \frac{\partial U}{\partial y}\right) = \frac{f_0}{\bar{\zeta}}$$

$$P < 0 \Longleftrightarrow R_i < \frac{f_0}{\bar{\zeta}}$$

If there no horizontal shear (Uy = 0), $R_i < 1$ for instability.

In frontal region, there can be no symmetric instability even $R_i < 1$

Symmetric instability trigged by latent heat in rainband.

In symmetric case, in quasi-geostrophic system, where $\zeta < f_0$ $R_i < 0$ is unstable.

No need to invoke the Kelvin Helmholtz instability ($R_i < 0.25$)

作業

作業1 解釋下列方程式代表意義，
那些變數是颱風主環流那些是次環流。

$$\frac{v^2}{r} + fv = \frac{\partial \phi}{\partial r}$$

$$\frac{\partial v}{\partial t} + u\left(f + \zeta\right) + w\frac{\partial v}{\partial z} = \mathbf{X}$$

$$\frac{\partial \phi}{\partial z} = \frac{g}{\theta_0}\theta$$

$$\frac{\partial \rho r u}{r \partial r} + \frac{\partial \rho w}{\partial z} = 0$$

$$\frac{\partial \theta}{\partial t} + u\frac{\partial \theta}{\partial r} + w\frac{\partial \theta}{\partial z} = \frac{\theta}{T}\frac{Q}{c_p}$$

作業2 解釋下列參數代表意義，
以及如何影響次環流形狀。

$$\rho A = \frac{g}{\theta_0}\frac{\partial \theta}{\partial z}$$

$$\rho B = -\frac{g}{\theta_0}\frac{\partial \theta}{\partial r} = -\left(f + \frac{2v}{r}\right)\frac{\partial v}{\partial z}$$

$$\rho C = \left(f + \frac{2v}{r}\right)\left(f + \frac{\partial r v}{r \partial r}\right)$$

作業3 解釋AC – B^2 代表意義，
說明次環流stiffness的特性。
次環流如何調整主環流的熱力風平衡。

作業4 推導次環流方程式，
了解其意義，了解為何要定義主與次環流，
彼此如何交互作用(準平衡)。[不用交]